

Dose selection for a phase III study evaluating gepotidacin (GSK2140944) in the treatment of uncomplicated urogenital gonorrhoea.

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Gepotidacin is a novel, first-in-class triazaacenaphthylene antibiotic that inhibits bacterial DNA replication by a distinct mechanism of action and is active against most strains of *Neisseria gonorrhoeae* (*N. gonorrhoeae*). Phase II data suggested higher exposures were needed for efficacy and to suppress resistance development. A translational approach using in vitro pharmacokinetic/pharmacodynamic (PK/PD) and clinical data was used to select a gepotidacin dose for a phase III study. In this narrative review of previously shown data, we summarise how a translational approach based on in vitro PK/PD and population PK modelling and simulation data was undertaken to select a dosing regimen for the ongoing phase III gepotidacin study in participants with uncomplicated urogenital gonorrhoea. For dose selection, prior in vitro minimum inhibitory concentrations (MICs) and PK/PD data were available. PK modelling was conducted to determine a dose that would limit plasma concentrations to less than 14 µg/mL (as concentrations above this are associated with QT prolongation and effects associated with acetylcholinesterase inhibition) while maintaining ≥90% probability of target attainment (PTA) for efficacy and resistance suppression against *N. gonorrhoeae* isolates with gepotidacin MICs ≤1 µg/mL. Two 3000 mg gepotidacin doses, administered 10-12 hours apart, resulted in PTA of ≥97.5% and ≥91.7% for gepotidacin MICs ≤1 µg/mL for the ratio of the area under the free drug plasma concentration-time curve over 24 hours to the MIC (fAUC₀₋₂₄/MIC) efficacy, and resistance suppression targets of 40 and 46, respectively, but limited the occurrence of maximum plasma concentrations ≥14 µg/mL. Two gepotidacin 3000 mg oral doses 10-12 hours apart provide ~2-fold higher systemic exposures, increase efficacy for higher gepotidacin MIC *N. gonorrhoeae* isolates, reduce resistance potential and limit plasma concentrations of potential safety concern, compared with higher doses.

Design of Two Phase III, Randomized, Multicenter Studies Comparing Gepotidacin with Nitrofurantoin for the Treatment of Uncomplicated Urinary Tract Infection in Female Participants.

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Uncomplicated urinary tract infections (uUTIs) are among the most common community-acquired infections for women worldwide. Treatment options are increasingly limited by antibiotic resistance; novel oral antibiotics are urgently needed. Gepotidacin is a novel, bactericidal, first-in-class triazaacenaphthylene antibiotic that inhibits bacterial deoxyribonucleic acid (DNA) replication by a distinct mechanism of action, which confers activity against most strains of target pathogens, such as *Escherichia coli* and *Staphylococcus saprophyticus*, including those resistant to current antibiotics. Here, we describe the designs of two near-identical phase III clinical trials (EAGLE-2 and EAGLE-3) evaluating gepotidacin for the treatment of uUTI. These are phase III, randomized, multicenter, parallel-group, double-blind, double-dummy, comparator-controlled, noninferiority studies, comparing the efficacy and safety of gepotidacin to nitrofurantoin in the treatment of uUTI. Eligible participants are women aged ≥ 12 years with ≥ 2 uUTI symptoms, randomized (1:1) to receive oral gepotidacin (1500 mg) plus placebo or nitrofurantoin (100 mg) plus placebo, twice daily for 5 days. The primary therapeutic endpoint is composite clinical and microbiological efficacy, with noninferiority comparisons made in individuals with a qualifying (≥ 10⁵ colony-forming units/mL urine) nitrofurantoin-susceptible uropathogen. These trials were designed in accordance with US Food and Drug Administration (2019) and European Medicines

Agency (2018) guidance, particularly the composite endpoint and microbiological evaluability requirements. Across the trials ~5000 participants are planned to be enrolled from >200 centers globally. Gepotidacin represents an important potential treatment option being evaluated to address the need for novel oral antibiotics to treat uUTI. These trials are registered at ClinicalTrials.gov (<https://clinicaltrials.gov/>) where the full protocols can be accessed under trial IDs: NCT04020341 (EAGLE-2) and NCT04187144 (EAGLE-3).

Novel Bacterial Topoisomerase Inhibitor Gepotidacin Demonstrates Absence of Fluoroquinolone-Like Arthropathy in Juvenile Rats.

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Fluoroquinolone use in children is limited due to its potential toxicity and negative effects on skeletal development, but the actual effects/risks of fluoroquinolones on bone growth and the mechanisms behind fluoroquinolone-driven arthropathy remain unknown. Gepotidacin is a novel, bactericidal, first-in-class triazaacenaphthylene antibiotic with a unique mechanism of action that is not anticipated to have the same risks to bone growth as those of fluoroquinolones. Gepotidacin is in phase III clinical development for uncomplicated urinary tract infections (ClinicalTrials.gov identifiers NCT04020341 and NCT04187144) and urogenital gonorrhea (ClinicalTrials.gov identifier NCT04010539) in adults and adolescents ≥ 12 years of age. To inform arthropathy and other potential toxicity risks of gepotidacin in pediatric studies, this nonclinical study assessed oral gepotidacin toxicity in juvenile rats from postnatal day (PND) 4 to PND 32/35 (approximately equivalent to human ages from newborn to 11 years), using both in-life assessments (tolerability, toxicity, and toxicokinetics) and terminal assessments (necropsy with macroscopic and microscopic skeletal femoral head and/or stifle joint examinations). Gepotidacin doses of ≤ 300 mg/kg of body weight/day were well tolerated from PND 4 to PND 21, and higher doses of $\leq 1,250$ mg/kg/day were well tolerated from PND 22 when the dose levels were escalated to maintain systemic exposure levels up to PND 35, with no observed treatment-related clinical signs, effects on mean body weight gain, or macroscopic findings on articular surfaces. A dose of 1,000 mg/kg/day was not tolerated during the dosing period from PND 4 to 21, with effects on body weight gain, fecal consistency, and body condition. Microscopic effects on articular surfaces were evaluated after 32 days of gepotidacin treatment at the highest tolerated dose. After 32 days of treatment with the highest tolerated gepotidacin dose of 300/1,250 mg/kg/day (systemic concentrations [area under the curve {AUC} values] of $93.7 \mu\text{g} \cdot \text{h/mL}$ [males] and $121 \mu\text{g} \cdot \text{h/mL}$ [females]), no skeletal effects on articular surfaces of the femoral head or stifle joint were observed. The absence of treatment-related clinical signs and arthropathy in juvenile rats provides evidence to support the potential future use of gepotidacin in children.

Gepotidacin (GSK2140944) In Vitro Activity against Gram-Positive and Gram-Negative Bacteria.

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Gepotidacin is a first-in-class, novel triazaacenaphthylene antibiotic that inhibits bacterial DNA replication and has in vitro activity against susceptible and drug-resistant pathogens. Reference in vitro methods were used to investigate the MICs and minimum bactericidal concentrations (MBCs) of gepotidacin and

comparator agents for *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Escherichia coli*. Gepotidacin in vitro activity was also evaluated by using time-kill kinetics and broth microdilution checkerboard methods for synergy testing and for postantibiotic and subinhibitory effects. The MIC₉₀ of gepotidacin for 50 *S. aureus* (including methicillin-resistant *S. aureus* [MRSA]) and 50 *S. pneumoniae* (including penicillin-nonsusceptible) isolates was 0.5 µg/ml, and for *E. coli* (n = 25 isolates), it was 4 µg/ml. Gepotidacin was bactericidal against *S. aureus*, *S. pneumoniae*, and *E. coli*, with MBC/MIC ratios of ≤4 against 98, 98, and 88% of the isolates tested, respectively. Time-kill curves indicated that the bactericidal activity of gepotidacin was observed at 4× or 10× MIC at 24 h for all of the isolates. *S. aureus* regrowth was observed in the presence of gepotidacin, and the resulting gepotidacin MICs were 2- to 128-fold higher than the baseline gepotidacin MICs. Checkerboard analysis of gepotidacin combined with other antimicrobials demonstrated no occurrences of antagonism with agents from multiple antimicrobial classes. The most common interaction when testing gepotidacin was indifference (fractional inhibitory concentration index of >0.5 to ≤4; 82.7% for Gram-positive isolates and 82.6% for Gram-negative isolates). The postantibiotic effect (PAE) of gepotidacin was short when it was tested against *S. aureus* (≤0.6 h against MRSA and MSSA), and the PAE-sub-MIC effect (SME) was extended (>8 h; three isolates at 0.5× MIC). The PAE of levofloxacin was modest (0.0 to 2.4 h), and the PAE-SME observed varied from 1.2 to >9 h at 0.5× MIC. These in vitro data indicate that gepotidacin is a bactericidal agent that exhibits a modest PAE and an extended PAE-SME against Gram-positive and -negative bacteria and merits further study for potential use in treating infections caused by these pathogens.

Cellular Pharmacokinetics and Intracellular Activity of Gepotidacin against *Staphylococcus aureus* Isolates with Different Resistance Phenotypes in Models of Cultured Phagocytic Cells.

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Gepotidacin (GSK2140944), a novel triazaacenaphthylene bacterial topoisomerase inhibitor, is currently in clinical development for the treatment of bacterial infections. This study examined in vitro its activity against intracellular *Staphylococcus aureus* (involved in the persistent character of skin and skin structure infections) by use of a pharmacodynamic model and in relation to cellular pharmacokinetics in phagocytic cells. Compared to oxacillin, vancomycin, linezolid, daptomycin, azithromycin, and moxifloxacin, gepotidacin was (i) more potent intracellularly (the apparent bacteriostatic concentration [C_s] was reached at an extracellular concentration about 0.7× its MIC and was not affected by mechanisms of resistance to the comparators) and (ii) caused a maximal reduction of the intracellular burden (maximum effect) of about -1.6 log₁₀ CFU (which was better than that caused by linezolid, macrolides, and daptomycin and similar to that caused by moxifloxacin). After 24 h of incubation of infected cells with antibiotics at 100× their MIC, the intracellular persisting fraction was 1% with the other drugs. The accumulation and efflux of gepotidacin in phagocytes were very fast (kin and kout, ~0.3 min⁻¹; the plateau was reached within 15 min) but modest (intracellular concentration-to-extracellular concentration ratio, ~1.6). In cell fractionation studies, about 40 to 60% of the drug was recovered in the soluble fraction and ~40% was associated with lysosomes in uninfected cells. In infected cells, about 20% of cell-associated gepotidacin was recovered in a sedimentable fraction that also contained bacteria. This study highlights the potential for further study of gepotidacin to fight infections where intracellular niches may play a determining role in bacterial persistence and relapses.

Microbiological Analysis from a Phase 2 Randomized Study in Adults Evaluating Single Oral Doses of Gepotidacin in the Treatment of Uncomplicated Urogenital Gonorrhea Caused by *Neisseria gonorrhoeae*.

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We evaluated microbiological correlates for the successful treatment of *Neisseria gonorrhoeae* isolates from a phase 2 study of gepotidacin, a novel triazaacenaphthylene antibacterial, for therapy of uncomplicated urogenital gonorrhea. Culture, susceptibility testing, genotypic characterization, and frequency of resistance (FoR) were performed for selected isolates. Microbiological success was defined as culture-confirmed eradication of *N. gonorrhoeae*. Against 69 baseline urogenital isolates, gepotidacin MICs ranged from ≤ 0.06 to $1 \mu\text{g/ml}$ ($\text{MIC}_{90} = 0.5 \mu\text{g/ml}$). For gepotidacin, the ratio of the area under the free-drug concentration-time curve to the MIC (fAUC/MIC) was associated with therapeutic success. Success was 100% (61/61) at fAUC/MICs of ≥ 48 and decreased to 63% (5/8) for fAUC/MICs of ≤ 25 . All 3 isolates from microbiological failures were ciprofloxacin resistant, had a baseline gepotidacin MIC of $1 \mu\text{g/ml}$, and carried a preexisting ParC D86N mutation, a critical residue for gepotidacin binding. In a test-of-cure analysis, the resistance to gepotidacin emerged in 2 isolates (MICs increased ≥ 32 -fold) with additional GyrA A92T mutations, also implicated in gepotidacin binding. Test-of-cure isolates had the same sequence type as the corresponding baseline isolates. For 5 selected baseline isolates, all carrying a ParC D86N mutation, the *in vitro* FoR to gepotidacin was low (10^{-9} to 10^{-10}); the resistant mutants had the same A92T mutation as the 2 isolates in which resistance emerged. Five participants with isolates harboring the ParC D86N mutation were treatment successes. In summary, fAUC/MICs of ≥ 48 predicted 100% microbiological success, including 3 isolates with the ParC D86N mutation (fAUC/MICs ≥ 97). Pharmacokinetic/pharmacodynamic determinations may help to evaluate new therapies for gonorrhea; further study of gepotidacin is warranted. (This study has been registered at ClinicalTrials.gov under identifier NCT02294682.).

Fgfr4 is required for effective muscle regeneration *in vivo*. Delineation of a MyoD-Tead2-Fgfr4 transcriptional pathway.

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Fgfr4 has been shown to be important for appropriate muscle development in chick limb buds; however, Fgfr4 null mice show no phenotype. Here, we show that staged induction of muscle regeneration in Fgfr4 null mice becomes highly abnormal at the time point when Fgfr4 is normally expressed. By 7 days of regeneration, differentiation of myotubes became poorly coordinated and delayed by both histology and embryonic myosin heavy chain staining. By 14 days much of the muscle was replaced by fat and calcifications. To begin to dissect the molecular pathways involving Fgfr4, we queried the promoter sequences for transcriptional factor binding sites and tested candidate regulators in a 27-time point regeneration series. The Fgfr4 promoter region contained a Tead protein binding site (M-CAT 5'-CATTCT-3'), and Tead2 showed induction during regeneration commensurate with Fgfr4 regulation. Co-transfection of Tead2 and Fgfr4 promoter reporter constructs into C2C12 myotubes showed Tead2 to activate Fgfr4, and mutation of the M-CAT motif in the Fgfr4 promoter abolished these effects. Immunostaining for Tead2 showed timed expression in myotube nuclei consistent with the mRNA data. Query of the expression timing and genomic sequences of Tead2 suggested direct regulation by MyoD, and consistent with this, MyoD directly bound to two strong E-boxes in the first intron of Tead2 by chromatin immunoprecipitation assay. Moreover, co-transfection of MyoD and Tead2 intron reporter

constructs into 10T1/2 cells activated reporter activity in a dose-dependent manner. This activation was greatly reduced when the two E-boxes were mutated. Our data suggest a novel MyoD-Tead2-Fgfr4 pathway important for effective muscle regeneration.

TEAD transcription factors are required for normal primary myoblast differentiation in vitro and muscle regeneration in vivo.

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The TEAD family of transcription factors (TEAD1-4) bind the MCAT element in the regulatory elements of both growth promoting and myogenic differentiation genes. Defining TEAD transcription factor function in myogenesis has proved elusive due to overlapping expression of family members and their functional redundancy. We show that silencing of either Tead1, Tead2 or Tead4 did not effect primary myoblast (PM) differentiation, but that their simultaneous knockdown strongly impaired differentiation. In contrast, Tead1 or Tead4 silencing impaired C2C12 differentiation showing their different contributions in PMs and C2C12 cells. Chromatin immunoprecipitation identified enhancers associated with myogenic genes bound by combinations of Tead4, Myod1 or Myog. Tead4 regulated distinct gene sets in C2C12 cells and PMs involving both activation of the myogenic program and repression of growth and signaling pathways. ChIP-seq from mature mouse muscle fibres in vivo identified a set of highly transcribed muscle cell-identity genes and sites bound by Tead1 and Tead4. Although inactivation of Tead4 in mature muscle fibres caused no obvious phenotype under normal conditions, notexin-induced muscle regeneration was delayed in Tead4 mutants suggesting an important role in myogenic differentiation in vivo. By combining knockdown in cell models in vitro with Tead4 inactivation in muscle in vivo, we provide the first comprehensive description of the specific and redundant roles of Tead factors in myogenic differentiation.

TEAD/TEF transcription factors utilize the activation domain of YAP65, a Src/Yes-associated protein localized in the cytoplasm.

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Mammals express four highly conserved TEAD/TEF transcription factors that bind the same DNA sequence, but serve different functions during development. TEAD-2/TEF-4 protein purified from mouse cells was associated predominantly with a novel TEAD-binding domain at the amino terminus of YAP65, a powerful transcriptional coactivator. YAP65 interacted specifically with the carboxyl terminus of all four TEAD proteins. Both this interaction and sequence-specific DNA binding by TEAD were required for transcriptional activation in mouse cells. Expression of YAP in lymphocytic cells that normally do not support TEAD-dependent transcription (e.g., MPC11) resulted in up to 300-fold induction of TEAD activity. Conversely, TEAD overexpression squelched YAP activity. Therefore, the carboxy-terminal acidic activation domain in YAP is the transcriptional activation domain for TEAD transcription factors. However, whereas TEAD was concentrated in the nucleus, excess YAP65 accumulated in the cytoplasm as a complex with the cytoplasmic localization protein, 14-3-3. Because TEAD-dependent transcription was limited by YAP65, and YAP65 also binds Src/Yes protein tyrosine kinases, we propose that YAP65 regulates TEAD-dependent transcription in response to mitogenic signals.

Insights into transcription enhancer factor 1 (TEF-1) activity from the solution structure of the TEA domain.

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Transcription enhancer factor 1 is essential for cardiac, skeletal, and smooth muscle development and uses its N-terminal TEA domain (TEAD) to bind M-CAT elements. Here, we present the first structure of TEAD and show that it is a three-helix bundle with a homeodomain fold. Structural data reveal how TEAD binds DNA. Using structure-function correlations, we find that the L1 loop is essential for cooperative loading of TEAD molecules on to tandemly duplicated M-CAT sites. Furthermore, using a microarray chip-based assay, we establish that known binding sites of the full-length protein are only a subset of DNA elements recognized by TEAD. Our results provide a model for understanding the regulation of genome-wide gene expression during development by TEA/ATTS family of transcription factors.
